

Lab 3, Act 2: Answer Key

Counting detections: binomial or Poisson?

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (the “plausible or not” part of 1a, 2e, and the group discussion) get sample answers or discussion points instead. All numbers are computed in R when this file is knitted.

Setting: the carnivore is confirmed present; a camera detects it on a given night with probability $p = 0.3$, independently from night to night, over $N = 8$ nights.

Part 1: Binomial by hand

1a.

Why is “number of nights with a detection, out of 8” a reasonable binomial random variable here? Name the assumptions the binomial distribution requires, and say briefly why each one seems plausible (or not) for this camera.

Answer. Four assumptions:

- **Fixed number of trials** ($N = 8$ nights): true by design.
- **Two outcomes per trial** (detection or not on each night): true by how the variable is defined.
- **Independent trials:** plausible but questionable. A storm lasting several nights, a camera malfunction, or an animal returning to a marked spot links one night to the next.
- **Constant p across trials:** also questionable. Detection can change with moon phase, weather, season, or the animal getting used to (or wary of) the camera.

Why. The first two hold by construction, so the substance of a good answer is in the last two.

Watch for. Listing the four assumptions with no judgment about which are shaky. Confusing “constant p ” with “ p is known.”

1b.

Using the binomial pmf, $P(X = k) = \binom{N}{k} p^k (1 - p)^{N-k}$, compute by hand the probability of exactly 3 detections in the 8 nights ($N = 8$, $k = 3$, $p = 0.3$). Show your work.

Answer. $P(X = 3) = \binom{8}{3} (0.3)^3 (0.7)^5 = 56 \times 0.027 \times 0.16807 = 0.2541$.

```
choose(8, 3) * 0.3^3 * 0.7^5
```

```
## [1] 0.2541218
```

```
dbinom(3, 8, 0.3) # check
```

```
## [1] 0.2541218
```

Why. $\binom{8}{3} = \frac{8!}{3!5!} = 56$ ways to choose which 3 nights had detections; each such sequence has probability $0.3^3 \times 0.7^5$.

Watch for. Leaving out $\binom{8}{3}$ (gives 0.0045, the probability of one particular sequence). Using $(0.7)^3(0.3)^5$. Arithmetic slips in $8!/(3!5!)$.

1c.

The mean of this distribution is $Np = 8 \times 0.3 = 2.4$, but the tallest bar (the mode) sits at a different whole number. Why can't the mode land exactly on the mean here? And what does the direction of the gap between them (mode below the mean, or above?) tell you about the shape of this distribution?

Answer. The mode is 2. It can't equal 2.4 because the number of detections must be a whole number, while the mean is an average and can fall between whole numbers. The mode sits below the mean, so the distribution is right-skewed: it's bounded at 0 on the left and has a longer tail toward higher counts, which pulls the mean up.

```
(0:8)[which.max(dbinom(0:8, 8, 0.3))] # mode
```

```
## [1] 2
```

```
8 * 0.3 # mean
```

```
## [1] 2.4
```

Why. In a right-skewed distribution, the few large values in the long tail pull the mean above the most common value.

Watch for. “Because the distribution is skewed” as the only answer to the first question (the whole-number reason comes first). Reading the skew backward (mode below mean means a right tail, not a left one).

Part 2: Comparing to Poisson

2a.

Look closely at both pairs of bar charts. In Scenario A, where specifically do the binomial and Poisson bars disagree most: the center, the spread, or the tails? Does Scenario B show the same kind of disagreement, a different kind, or none at all?

Answer. Scenario A: the spread. The binomial is taller in the middle (2 to 4 detections); the Poisson puts more probability in both tails (0 and 1 detections, and 5 or more). Scenario B: almost no disagreement anywhere.

```
round(rbind(binom_A = dbinom(0:8, 8, 0.3),
            binom_B = dbinom(0:8, 100, 0.024),
            poisson = dpois(0:8, 2.4)), 3)
```

```
##           [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9]
## binom_A 0.058 0.198 0.296 0.254 0.136 0.047 0.010 0.001 0.000
## binom_B 0.088 0.217 0.264 0.212 0.126 0.060 0.023 0.008 0.002
## poisson 0.091 0.218 0.261 0.209 0.125 0.060 0.024 0.008 0.002
```

Why. The Poisson is wider than Binomial(8, 0.3) even though both have mean 2.4 (2b shows why). Binomial(100, 0.024) has nearly the same spread as the Poisson.

Watch for. “The Poisson has a fatter right tail” alone; it's also higher at 0 and 1. “The center differs”: both means are 2.4.

2b.

Compute the variance of each distribution by hand, using $\text{Var} = Np(1 - p)$ for the binomial and $\text{Var} = \lambda$ for the Poisson.

Answer.

- Scenario A binomial: $8 \times 0.3 \times 0.7 = 1.68$.
- Scenario B binomial: $100 \times 0.024 \times 0.976 = 2.3424$.
- Poisson (both scenarios): $\lambda = 2.4$.

Why. Direct substitution. The binomial variance equals the Poisson variance times $(1 - p)$, so the two match only when $1 - p$ is close to 1.

Watch for. Using p instead of $1 - p$ in the binomial formula. Computing only Scenario A.

2c.

Does the pattern of variances you just computed explain what you saw by eye in 2a?

Answer. Yes. In Scenario A the Poisson's variance (2.4) is about 40% larger than the binomial's (1.68), matching its wider spread. In Scenario B the variances are nearly equal (2.40 vs. 2.34), matching the near-identical bars.

Why. With the same mean, the variance is what controls the width.

Watch for. "Yes" with no reference to the numbers.

2d.

Finish this rule of thumb in your own words: "*A Poisson distribution is a good stand-in for a binomial distribution when...*"

Answer. "...the number of trials is large and the probability per trial is small," so that $1 - p \approx 1$ and the binomial variance $Np(1 - p)$ is close to the Poisson's $\lambda = Np$.

Why. Scenario B (large N , small p) matched; Scenario A (small N , $p = 0.3$) didn't.

Watch for. "When the means are equal" (they were equal in both scenarios). "Always" based on Scenario B alone. Mentioning only large N or only small p ; both are needed.

2e.

Back in the original 8-night scenario ($p = 0.3$): name one realistic reason the constant, independent per-night detection probability assumption from 1a might be violated in the field.

Sample answer. Any one realistic mechanism, for example:

- A multi-night storm suppresses detections on several nights in a row (breaks independence).
- Moon phase or season changes the animal's activity (breaks constant p).
- The animal learns to avoid the camera, or is drawn to the site by scent (changes p over the deployment).

Why. Each of these makes a night's detection depend on something other than a single fixed p .

Watch for. Reasons that change whether the animal is present (the Act assumes it is). Restating the assumption without a mechanism.

Group discussion and revision

Compare your variances and rule of thumb from 2d with your group. Record any changes, new insights, and muddiest points.

Discussion points. No fixed answer. Useful reflections connect the variance formulas in 2b to the picture in 2a, or note why matching means alone doesn't make two distributions interchangeable.